

Service@home ontology validation

Dear participant,

Thanks for participating in our survey. Your contribution is very important to improve our work.

This Survey is part of a postdoctoral project that aims to specify an ontology for the service robots domain, named Service@Home ontology. The project is a collaboration between the State University of Bahia (UNEB) and the Federal Technological University of Paraná (UTFPR).

The main goal of this survey is to evaluate the Service@Home ontology terminologies with respect to clarity, legibility, coherence and completeness from the perception of Robocup@home team members.

Questionnaire is voluntary, has 16 questions, as well as some demographic data, and takes approximately 20 minutes to complete (7 pages of multiple-choice questions). We guarantee that this research follows a high academic standard and your data will be kept anonymous. We will not associate your email address with your responses and will use it exclusively to contact you in the next steps, if necessary.

By clicking the "Next" button, you agree to participate in this survey. Consent also indicates that you agree that all information collected individually may be used by researchers, so that your personal identity is preserved and protected. Such use will include presentations at scientific or professional meetings, publication in scientific journals, sharing anonymous information with other researchers to verify the accuracy of study results and for future approved research that has the potential to advance knowledge.

For more information, do not hesitate to contact us via the email below:

- Ana Patrícia Magalhães (anapatriciamagalhaes@gmail.com), Ph.D, State University of Bahia (UNEB)

This questionnaire will be available for responses until **30/06/2024**.

We thank you for your attention, participation and support.

* Indica uma pergunta obrigatória

DEMOGRAPHIC DATA - Step 2/8 (5 questions)

1. What is your Email address? *

2. What is your RoboCup@Home team name? *

3. What is your instruction level? *

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- ☐ Undergraduate student
- ☐ Post graduate student (master degree)
- ☐ Post graduate student (doctorial degree)
- ☐ Master degree
- ☐ Doctorate degree

4. What is your area? *

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- ☐ Computing
- ☐ Electrical Engineering
- ☐ Mechanical Engineering
- ☐ Mechatronic
- ☐ Other

5. In you answer "other", please specify which.

6. What is your experience level working with service robot? *

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- ☐ Beginner (up to 2 years)
- ☐ Intermediary (up to 4 years)
- ☐ Advanced (more than 4 years)

7. What is your role on the @home team? *

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- ☐ Team member (students involved in the robot development)
- ☐ Captain (student that represents the team)
- ☐ Team leader (Professor that coordinate the team)

INTRODUCTION - Step 3/8

The ontology concepts are organized into three categories:

- Concepts related to the **robot structure** and capabilities;
- Concepts related to the **environment**, where the robot will operate;
- Concepts related to the **tasks** that robot will execute.

Next sections present examples and ontology concepts according to the above categories.

Please take note of the time spent to complete this survey.

ROBOT STRUCTURE - Step 4/8 (6 questions)

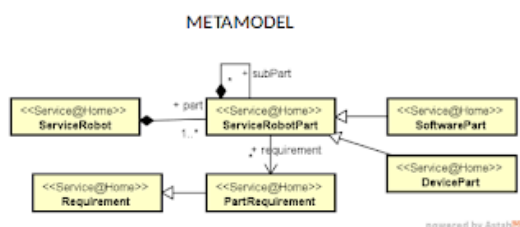
8. The following pictures present an example of a robot (blue box) and the ontology concepts used in it, expressed in natural language (yellow box) and in a metamodel (bottom). *

EXAMPLE

Hello, my name is BILL!
I 'm here to help you doing your domestic tasks.

I have a camera, a speaker, a computer, among other hardware. I also have a person detection algorithm, a speech program, and a navigation system.

My camera has a resolution of 24MP.



CONCEPTS

ServiceRobot: is a fully autonomous Robot, for personal or professional use, that performs useful tasks for humans or equipment. It may involve human interaction, but does not allow human intervention.

Human Interaction means information and action exchanges between human and robot. Human intervention means external (remote or in-person) human control.

Personal use means assistance in a house, e.g. are cooking or cleaning.

Professional use means, assistance in a non-structured environment, e.g., serving dinner in a restaurant.

ServiceRobotPart: represents the parts of a ServiceRobot. A part can be composed of other parts (named subpart).

SoftwarePart: represents the software components, e.g., software functions.

DevicePart: represents the hardware parts.

Requirement: represents a proposition that must be fulfilled.

PartRequirement: is the requirement of a part.

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	ServiceRobot	DevicePart	SoftwarePart	PartRequirement
Bill	☐	☐	☐	☐
Computer	☐	☐	☐	☐
Person detection algorithm	☐	☐	☐	☐
Camera	☐	☐	☐	☐
Resolution of 24MP	☐	☐	☐	☐
Navigation system	☐	☐	☐	☐

9. How compatible are the following concepts with the vocabulary used by the community? (mark 1-incompatible and 5 - fully compatible) *

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	1	2	3	4	5
ServiceRobot	☐	☐	☐	☐	☐
DevicePart	☐	☐	☐	☐	☐
SoftwarePart	☐	☐	☐	☐	☐
Requirement	☐	☐	☐	☐	☐

10. Please suggest another terminology for each concept that you mark 1-3 above.
(e.g. ServiceRobot -> YourSuggestion)

11. The following pictures present an example of what BILL can do (blue box) and the ontology concepts used in it, expressed in natural language (yellow box) and in a metamodel (bottom). *

Analyze the picture, then match the terms used in the example with their respective ontology concept.

EXAMPLE

- BILL can
 - Navigate
 - Recognize face
 - Recognize object
 - Interact with humans using voice
- To recognize a face it is necessary first to capture a picture to then identify the face in the picture.
- To recognize a face or an object, the algorithm needs an image of at least 20MP. BILL's camera captures 24MP images, thus, it may be used along with such recognition algorithms.
- BILL uses the person detection algorithm to identify one person's face based on an image captured by its camera.

CONCEPTS

Capability: is the combination of behaviors, robot parts, and organization that delivers a specified outcome. A capability specifies **WHAT** the robot can deliver without implementation.

ServiceRobotBehavior: represents a capability realization according to a specific implementation strategy. It orchestrates the set of **PrimitiveActions** necessary for a capability implementation. **PrimitiveActions** can be orchestrated in parallel or sequentially.

PrimitiveAction: consists of using **ServiceRobotParts** to produce a specific result. A **PrimitiveAction** is a behavior stage which **CANNOT** be composed by other **PrimitiveAction**.

Sensing: related to sensing and capturing data from the environment.

Cognition: related to data processing.

Actuating: related to act and modify the environment.

Contract: establishes an agreement between a **PrimitiveAction** and a **ServiceRobotPart**

generated by Astah

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	Capability	ServiceRobotBehavior	PrimitiveAction	Contract
Navigate	☐	☐	☐	☐
Recognize face	☐	☐	☐	☐
Description of the actions performed to recognize a face	☐	☐	☐	☐
Capture picture	☐	☐	☐	☐
Identify face	☐	☐	☐	☐
The agreement between capture picture and	☐	☐	☐	☐

the camera
picture and
the camera

12. How compatible are the following concepts with the vocabulary used by the community? (mark 1-incompatible and 5 - fully compatible) *

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	1	2	3	4	5
Capability	☐	☐	☐	☐	☐
ServiceRobotBehavior	☐	☐	☐	☐	☐
PrimitiveAction	☐	☐	☐	☐	☐
Contract	☐	☐	☐	☐	☐

13. Please suggest another terminology for each concept that you mark 1-3 above.
(e.g. Capability -> YourSuggestion)

ENVIRONMENT - Step 5/8 (6 questions)

14. The following pictures present an example of an environment where BILL can operate (blue box) and the ontology concepts used in it, expressed in natural language (yellow box) and in a metamodel (bottom).

Analyze the picture, then match the terms used in the example with their respective ontology concept.

EXAMPLE

Bill will operate in an Arena of Robocup@Home at Robotica 2025 in Salvador/Brazil! The arena has a living room, a dining room, a kitchen and a bedroom.

The living room has one table and four chairs. A flower jar lies on the table.

In the kitchen, there are a refrigerator, a shelf and some objects such as a bottle of water, an apple, and a plate. These objects are located on the shelf and will be used in the object recognition task.



CONCEPTS

Scenario: is a physical environment where a **ServiceRobot** performs tasks. It contains at least one **Area**. A **Scenario** can be of two types: **DomesticScenario** or **PublicScenario**.

DomesticScenario: is a typical domestic setting.

PublicScenario: is a non-structured place, e.g., a restaurant.

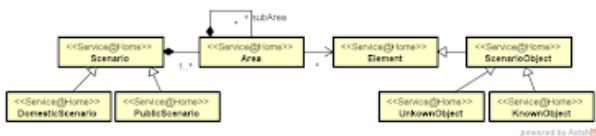
Area: represents the parts of a Scenario. It may be composed of another **Area** (**Subarea**).

Element: is anything or anybody inside an **Area**. An **Element** has a **Pose** and can be of one of two types: **ScenarioObject** or **Agent**.

ScenarioObject: represents things arranged in an **Area**. A **ScenarioObject** has a name, a description, a category (e.g. furniture, decoration, tableware and bags), color, weight, shape, can be deformable, has a texture and a material.

KnownObject: is a **ScenarioObject** previously known, i.e., trained, by the robot.

UnknownObject: is a **ScenarioObject** not known, i.e., not trained, by the robot.



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	DomesticScenario	PublicScenario	Area	KnownObject	UnkownObje
Arena of Robocup@Home	☐	☐	☐	☐	☐
Dining room	☐	☐	☐	☐	☐
Kitchen	☐	☐	☐	☐	☐
Table	☐	☐	☐	☐	☐
Flower jar	☐	☐	☐	☐	☐
Bottle of water	☐	☐	☐	☐	☐
Apple	☐	☐	☐	☐	☐

15. How compatible are the following concepts with the vocabulary used by the community? (mark 1-incompatible and 5 - fully compatible)

*

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	1	2	3	4	5
DomesticScenario	☐	☐	☐	☐	☐
PublicSpace	☐	☐	☐	☐	☐
Area	☐	☐	☐	☐	☐
UnknownObject	☐	☐	☐	☐	☐
KnownObject	☐	☐	☐	☐	☐

16. Please suggest another terminology for each concept that you mark 1-3 above.
(e.g. DomesticScenario -> YourSuggestion)

17. The following pictures present an example of an environment where BILL can operate (blue box) and the ontology concepts used in it, expressed in natural language (yellow box) and in a metamodel (bottom).

Analyze the picture, then match the terms used in the example with their respective ontology concept.

EXAMPLE

In the kitchen there are a housekeeper and a dog.

João and Maria are in the living room. Maria is the robot operator.

The table is located in a specific coordinate (x=1, y=2, z=1).

When BILL detects the bottle of the water, the algorithm draws a rectangle around the bottle indicating its position.

João is near the table.

CONCEPTS

Agent: is Something or someone that can act on its own and produce changes in the world. It may interact with the service robot.

Person: represents an unknown human participating in the scenario. The **ServiceRobot** can recognize that this **Agent** is a human, but it does not have information about him.

KnownPerson: represents any known human, i.e. a **Person** that the **ServiceRobot** can recognize, it has information about this person, i.e. its face, clothes color or any other attribute that makes it recognizable. **KnownPerson** is associated to one or more **Roles**.

Role: defines the purpose/position of a **Person**, e.g. a resident of a house, an elderly person, a nurse caring for an elderly person. Each **Role** is associated to one or more **PrivilegeKind**.

PrivilegeKind (Service@Home): defines the rights a person can have when interacting with the serviceRobot.

3DPoint: is a 3D point in a position coordinate system indicating a specific location.

BoundingBox: a set of position points in a given coordinate system, delimited by a rectangle surrounding an **Element**.

RelativeRegion: indicates a qualitative position about a referenced **Element**. Possible qualitative positions are pre-defined, e.g. near, next, over, under.

* Classes in gray were imported from other ontologies

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	Agent	Person	KnownPerson	Role	3DPoint	BoundingBox	RelativeRe
Housekeeper	☐	☐	☐	☐	☐	☐	☐
Dog	☐	☐	☐	☐	☐	☐	☐
João	☐	☐	☐	☐	☐	☐	☐
Maria	☐	☐	☐	☐	☐	☐	☐
Robot operator	☐	☐	☐	☐	☐	☐	☐
Coordinate (x=1,y=2,z=1)	☐	☐	☐	☐	☐	☐	☐
A rectangle	☐	☐	☐	☐	☐	☐	☐
Near the table	☐	☐	☐	☐	☐	☐	☐

18. How compatible are the following concepts with the vocabulary used by the community? (mark 1-incompatible and 5 - fully compatible)

*

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	1	2	3	4	5
Agent	☐	☐	☐	☐	☐
Person	☐	☐	☐	☐	☐
KnownPerson	☐	☐	☐	☐	☐
Role	☐	☐	☐	☐	☐
3DPoint	☐	☐	☐	☐	☐
BoundingBox	☐	☐	☐	☐	☐
RelativeRegion	☐	☐	☐	☐	☐

19. Please suggest another terminology for each concept that you mark 1-3 above.
(e.g. Person -> YourSuggestion)

TASK - Step 6/8 (3 questions)

20. The following pictures present an example of Task executed by BILL (blue box) and the ontology concepts used in it, expressed in natural language (yellow box) and in a metamodel (bottom). *

Analyze the picture, then match the terms used in the example with their respective ontology concept.

- BILL performed some tasks during the RoboCup@Home in Salvador.
- One of these tasks was the Personal Recognition, which main goal is "The robot has to identify the Operator within a crowd and state information about the Operator and the crowd."
- To perform this task Bill followed the algorithm:
 1. Add new face, which aims to learn a new face. It involves:
 - a. Give Information, where Bill directs the operator to stand in front of it
 - b. Learn Face, where Bill memorizes the operator
 2. Recognize, which aims to recognize the operator. It involves:
 - a. Give Information, where Bill asks the operator to move to the crowd
 - b. Spin, where Bill turns 180 degrees
 - c. Recognize Face, where Bill recognizes the operator's face
 - d. Count People, where Bill counts the number of people in the scene
 3. Give results, which aims to give the results of the recognition. It involves:
 - a. Give Information, where Bill describes the recognition
 - b. Walk To, where Bill stays in front of the operator
 - c. Look To, where Bill looks to the operator

CONCEPTS

Task: represents the work performed by the **ServiceRobot** when serving a human. A **Task** has a name, a description, main goals (at least one), and may have optional goals. **Tasks** are performed in a set of steps organized as an **ExecutionPlan**.

ExecutionPlan: is an arrangement of **Steps** for **Task** execution. In this arrangement, **Steps** can be orchestrated in parallel or sequentially using, for example, behavior tree or UML state machine.

Step: represents a stage of the **ExecutionPlan** with a particular purpose. There are two types of Steps: **ComposedStep** and **SingleStep**.

ComposedStep: is a **Step** composed of other **Steps**. The composition of **Steps** must end in a **SingleStep**.

SingleStep: is a **Step** that can not be composed of other steps (e.g. considering a tree, steps are the leaves of the tree). It refers to a **Capacity**.

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	Task	Execution plan	Composed step	Single step
PersonalRecognition	☐	☐	☐	☐
Algorithm	☐	☐	☐	☐
Add new face	☐	☐	☐	☐
Give results	☐	☐	☐	☐
Give information	☐	☐	☐	☐
Learn face	☐	☐	☐	☐
Spin	☐	☐	☐	☐
LookTo	☐	☐	☐	☐

21. How compatible are the following concepts with the vocabulary used by the community? (mark 1-incompatible and 5 - fully compatible) *

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	1	2	3	4	5
Task	☐	☐	☐	☐	☐
ExecutionPlan	☐	☐	☐	☐	☐
ComposedStep	☐	☐	☐	☐	☐
SingleStep	☐	☐	☐	☐	☐

22. Please suggest another terminology for each concept that you mark 1-3 above. (e.g. Task -> YourSuggestion)

FINAL COMMENTS - Step 7/8

23. Do you have any more suggestions / comments to contribute with our ontology?

24. How much time (in minutes) did you spend in this survey? *

INFORMATION ABOUT OUR PROJECT - Step 8/8

Thanks for your contribution!

The preliminary version of the Service@Home methodology is available at:

<https://drive.google.com/file/d/1C2y-bT-TiRaRMGAuDWMzaGW39LPod0kM/view?usp=sharing>

This is a "preliminary" version, which will be published, together with the evaluations and feedback obtained with this survey, in a scientific vehicle to be defined.

Those who completed the survey will be informed where and when it is published!

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